WT-EXPERIMENT: 06

Topic: Radar Detection System.

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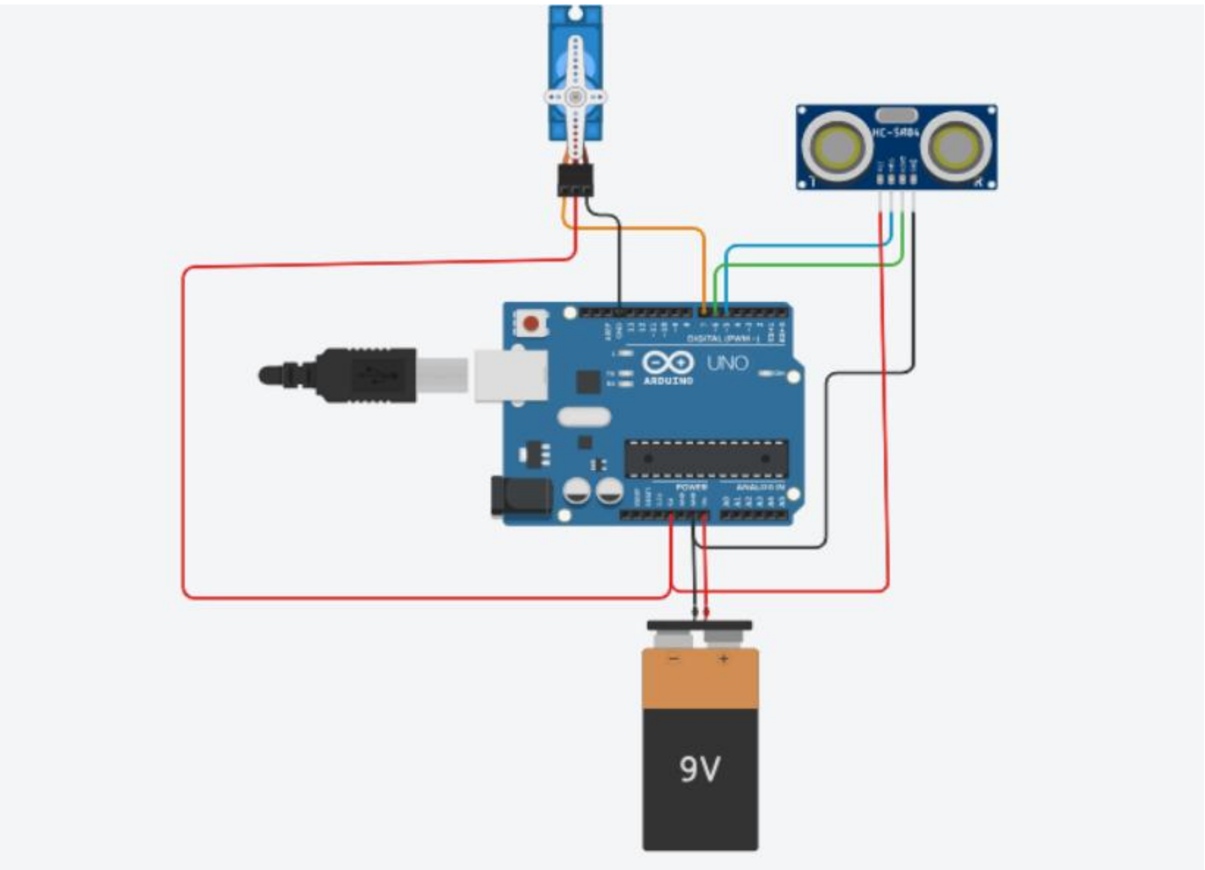
Aim: To design and implement a basic radar detection system using Arduino UNO, an ultrasonic sensor (HC-SR04), and a servo motor to simulate angular scanning for obstacle detection.

Introduction:

Radar systems are widely used in fields such as aviation, defense, automotive safety, and robotics for detecting objects and measuring distances. Inspired by the basic working principle of radar, this project aims to simulate a simplified radar detection system using an Arduino UNO, an HC-SR04 ultrasonic sensor, and a servo motor. Instead of using complex radio waves, this system uses ultrasonic waves to detect the presence of nearby objects and measure the distance to them.

The ultrasonic sensor is mounted on a servo motor, which allows it to rotate in an arc from 0° to 180°, mimicking the sweeping motion of a radar dish. As the servo moves, the sensor sends out ultrasonic pulses and listens for echoes. Based on the time taken for the echo to return, the Arduino calculates the distance of the object in front of the sensor. These readings, along with the corresponding angles, are then displayed in real time on the Serial Monitor of the Arduino IDE.



Components Used:

- Arduino UNO
- HC-SR04 Ultrasonic Sensor
- Servo Motor (SG90 or similar)
- 9V Battery with clip
- Jumper wires
- Breadboard (optional)

Connection of Components with Arduino UNO- Radar Detection System:

Component	Pin on Component	Connected to Arduino UNO
HC-SR04 Ultrasonic Sensor	VCC	5V
	GND	GND
	Trig	Digital Pin 10
	Echo	Digital Pin 11
Servo Motor	Signal (Orange/Yellow)	Digital Pin 9
	VCC (Red)	5V
	GND (Brown/Black)	GND
Power Supply	9V Battery (via clip)	VIN and GND on Arduino
USB Cable (Optional)	USB Port	For code upload and monitoring

Code for Connection:

```
#include <Servo.h>

#define trigPin 10

#define echoPin 11

Servo myServo;

void setup() {

  Serial.begin(9600);

  myServo.attach(9);

  pinMode(trigPin, OUTPUT);
```

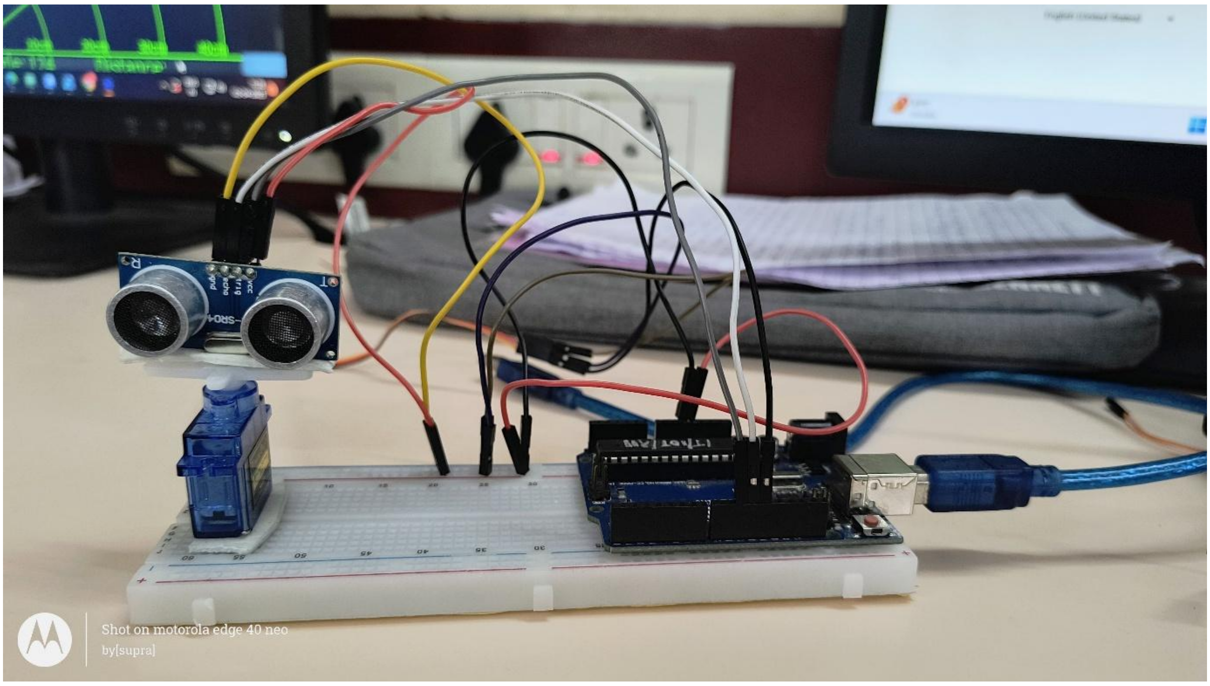
```
  pinMode(echoPin, INPUT);
}
void loop() {
  for (int angle = 0; angle <= 180; angle += 10) {
    myServo.write(angle);
    delay(300);
    float distance = measureDistance();
    Serial.print("Angle: ");
    Serial.print(angle);
    Serial.print("° | Distance: ");
    Serial.print(distance);
    Serial.println(" cm");
  }
  for (int angle = 180; angle >= 0; angle -= 10) {
    myServo.write(angle);
    delay(300);
    float distance = measureDistance();
    Serial.print("Angle: ");
    Serial.print(angle);
    Serial.print("° | Distance: ");
    Serial.print(distance);
    Serial.println(" cm");
  }
}
float measureDistance() {
```

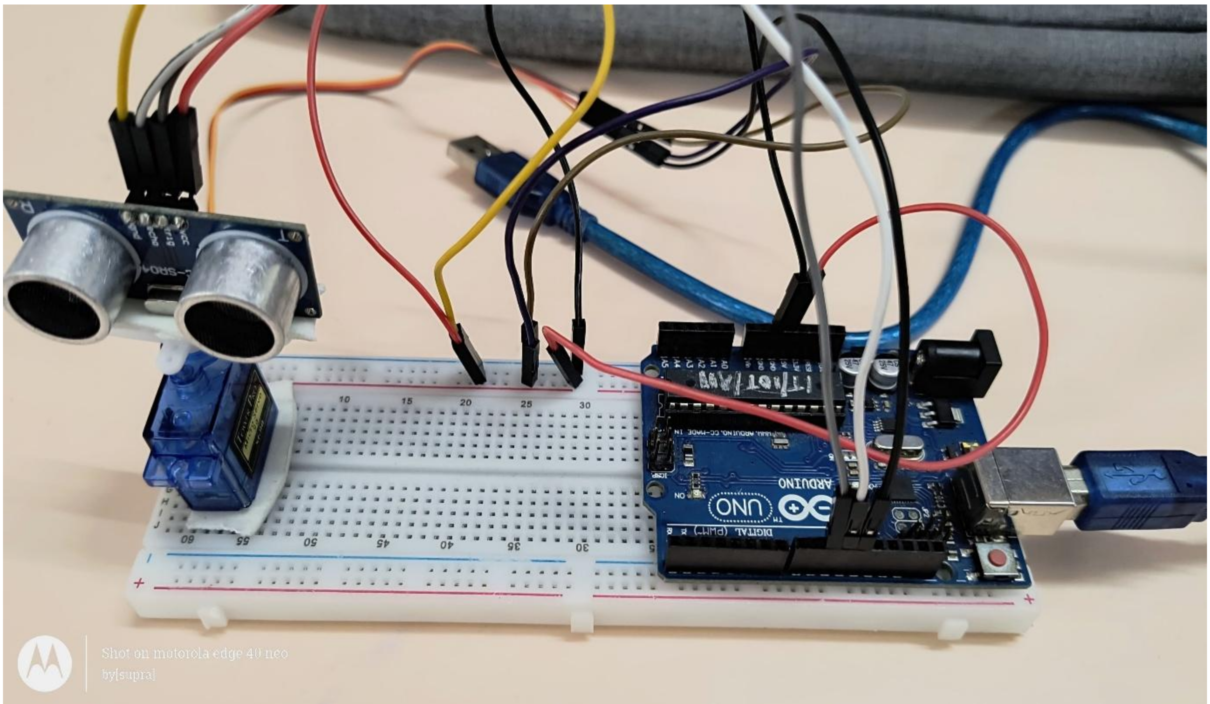
```
  digitalWrite(trigPin, LOW);

  delayMicroseconds(2);

  digitalWrite(trigPin, HIGH);

  delayMicroseconds(10);

  digitalWrite(trigPin, LOW);

  long duration = pulseIn(echoPin, HIGH);

  float distance = duration * 0.034 / 2;

  return distance;

}
```

Libraries Used:

1. Servo.h: Allows control over the servo motor to rotate between 0–180° angles.
2. No additional libraries are needed as the ultrasonic sensor uses simple digital pins.

Implementation:





1. Connect the components according to the wiring diagram (as shown in your image).
2. Upload the code to the Arduino UNO via USB.
3. Open the Serial Monitor (set baud rate to 9600).
4. The servo rotates from 0° to 180° and back, measuring distances at each step.
5. Distance data is printed to the Serial Monitor, emulating radar behavior.

Conclusion:

This project successfully demonstrates a simple radar-like object detection system using Arduino. The system scans the environment by rotating an ultrasonic sensor on a servo motor and prints the measured distances to the Serial Monitor. It's a minimal yet effective simulation of how radar systems work in real applications like obstacle avoidance, robotics, and surveillance.